



Tower Simulation System Enhancement 2 Challenge-Based Acquisition

Request for Concept Paper

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1. Introduction

1.1 Background

The Federal Aviation Administration (FAA) operates a complex air traffic control system that relies on highly trained controllers to manage aircraft movements safely and efficiently. The FAA uses high-fidelity air traffic control simulators, called Tower Simulation Systems (TSSs), to train and evaluate controllers in performing air traffic control operations. TSSs enable the FAA to design training scenarios to meet the FAA's training objectives and allow controllers to practice air traffic control procedures in a highly realistic virtual environment without risk to National Airspace System (NAS) operations. The FAA currently operates 111 TSSs to meet the training needs of controllers stationed at all 264 FAA-operated air traffic control towers (ATCTs) throughout the NAS.

The FAA has an urgent need to expand the controller workforce to address current and anticipated staffing shortages. TSSs, which can accelerate controller training and reduce the time required for certification, will play an important role in helping the FAA achieve its controller staffing goals. However, there are 173 ATCTs in the NAS that do not have a TSS on site. Controllers at these ATCTs must travel to other facilities for simulation training, incurring costs and requiring time for travel that exacerbates existing controller staffing shortages.

The FAA recognizes the need for additional TSSs. Furthermore, the FAA Reauthorization Act of 2024 mandated that the FAA improve controllers' access to high-fidelity simulators by May 2028. Under the TSS E2 acquisition, the FAA is seeking an enhanced TSS solution that will be deployed to 102 training sites by May 2028 and TSS operation and maintenance services through the end of the period of performance.

In May 2025, the FAA announced the Brand-New Air Traffic Control System (BNATCS) plan, a comprehensive effort to transform the United States' air traffic control system. To lead and manage the BNATCS effort across the FAA, the FAA awarded a prime integrator contract to Peraton.

Due to the TSS's importance to achieving the FAA's controller staffing goals, the FAA has designated the TSS program as part of the BNATCS portfolio.

1.2 Purpose

The FAA is issuing this Request for Concept Paper (RCP) to initiate the TSS Enhancement 2 (E2) Challenge-Based Acquisition (ChBA). Under the ChBA framework, the FAA or its designated agent may conduct a multi-phase evaluation process involving challenges, which could include demonstrations, prototypes, presentations, and proposals, culminating in the selection of a complete TSS E2 solution. The purpose of this RCP is to identify the offerors eligible to participate in these future phases. In later phases, the FAA or its designated agent may award one or more contracts or subcontracts to procure the hardware, software, and services necessary to implement the selected solution. If a viable solution is developed, the FAA or its designated agent may proceed with one or more contracts or subcontracts to implement that solution.

This RCP is not a Screening Information Request (SIR) or Request for Offers (RFO). The responses to

this announcement will not be construed as a commitment or a promise to contract with the FAA. The FAA is not seeking or accepting unsolicited proposals. Any costs associated with the response to this announcement are solely at the interested respondent's expense. Material will not be returned.

1.3 Attachments

The scope of work, objectives, and tasks needed to modernize the TSS are described in the following attachments:

- TSS E2 Statement of Objectives (SOO)
- TSS E2 System Specification* – Defines the requirements for the current TSS System. The TSS E2 System must meet the requirements in the TSS System Specification. However, offerors are permitted to propose changes to requirements in the TSS System Specification with an explanation of the benefits of the proposed change.
- Representative TSS Hardware Components* – Lists the hardware components of the current TSS.
- Representative TSS RPO Hardware Components* – Lists the hardware components of the current TSS remote pilot operator (RPO) workstation.
- TSS E2 Site List* – Indicates the facilities to which the FAA would like to deploy TSSs under the TSS E2 effort.
- List of Visual Databases and Number of Associated Training Scenarios* – Lists the visual databases and training scenarios that must be prepared for use by the TSS E2 System.
- TSS E2 Concept of Operations (ConOps)*
- TSS E2 Sample Training Scenario in current format*
- TSS E2 Sample Visual Database in current format*
- TSS E2 Aircraft and Vehicle List*

*This file can be accessed at <https://faa-aoc.box.com/s/zlev2xhzssgdigj0okp5ny95d89n1dm0>.

2. TSS E2 Challenge-Based Acquisition Overview

The purpose of this section is to provide an overview of the phases of the TSS E2 ChBA.

- **Phase I – Concept Paper:** The FAA issues the RCP. Responses may be provided to the FAA or its designated agent. Upon review of the responses, the FAA or its designated agent will invite offerors to participate in future phases.
- **Phase II – Proof of Concept:** The FAA or its designated agent will provide offerors with up to 60 days to prepare a proof of concept against scripts and use cases. This will culminate in a proof-of-concept demonstration and a FAA- or FAA-designated-agent-facilitated question and answer period. The offerors that best address the scripts, use cases provided, and the

associated evaluation criteria will be invited to future phases, if required, or to enter into a subcontract agreement.

3. Instructions to Offerors

3.1 Minimum Acceptability

The FAA requests that offerors' submissions conform to the following requirements:

- Contain a one-page cover sheet with the following information: Company name; Name of proposed TSS solution; Date of submittal; and Primary point of contact, including name, company, title, address, telephone number, and email address
- Comply with the formatting requirements in section 3.2
- Be submitted according to the instructions in section 3.4

3.2 Formatting

Concept papers must adhere to the following formatting requirements:

- The font must be Times New Roman 12 point (or larger) single-spaced. Smaller fonts may be used for figures and tables but must be clearly legible.
- Pages must be single-sided, 8.5 by 11 inches, with at least 1-inch margins on all sides (top, bottom, left, and right).
- Page limit is twenty-five (25) total pages excluding the cover sheet, table of contents, and the ROM Estimate. Page limitations must not be circumvented by including inserted text boxes, pop-ups or internet links to additional information. Such inclusions are not acceptable and will not be considered as part of the response to this RCP.

3.3 Concept Paper

As part of Phase I of the Challenge-Based Acquisition, offerors are asked to describe how they would meet the objectives in the TSS SOO. Offerors are asked to describe how they would specifically address the operational concept, technical, implementation and deployment, and operations and maintenance model for delivering a TSS solution. Offerors must include the following sections in their response:

- I. Technical Approach and Solution Architecture
- II. Implementation Approach
- III. Operational Approach
- IV. ROM Estimate

3.3.1 Technical Approach and Solution Architecture

Offerors must describe their proposed solution and how it meets objectives 1-7 of the SOO. Offerors must identify the components of their solution and describe their level of maturity with respect to meeting the objectives. Offerors must identify relevant risks and their proposed approach for managing them.

3.3.2 Implementation Approach

Offerors must describe their approach and timeline to delivering a complete solution by May 2028 to meet objective 14 of the SOO. Offerors must describe their approach to acquiring and deploying the software, hardware, personnel, and facilities used to produce, deliver, operate, and maintain the solution. Offerors must identify relevant risks and their proposed approach for managing them.

3.3.3 Operational Approach

Offerors must describe their approach to meeting objectives 8-13 of the SOO. Offerors must describe the roles of team members, facilities, and support resources. Offerors must identify relevant risks and their proposed approach for managing them.

3.3.4 ROM Estimate

Offerors must provide a rough order of magnitude (ROM) estimate of the total cost of their proposed TSS E2 solution in an offeror-developed format that includes the cost to develop and deploy TSSs to 102 designated sites covering the full deployment period through May 2028 and for one (1) additional year of operational and maintenance services. The ROM estimate must address each of the cost elements separately.

Offerors must provide a narrative describing their basis of estimate (BOE) which contains clear traceability between each BOE and each cost element. The BOE supporting data must explain what is included in the price of each element and provide traceability between the priced services and the technical approach, business model, and implementation plan. The BOE for each cost element must contain sufficient detail to allow the evaluators to assess the cost and price realism, balance risk, completeness, consistency, and traceability, as needed. Offerors must clearly identify and explain any cost elements that have interdependencies with other cost elements.

3.4 Submission Instructions

An offeror that includes in its concept paper data that it does not want to be disclosed to the public for any purpose or used by the FAA except for consideration and selection purposes must mark the information "Proprietary."

Responses are due by 2:00PM Eastern Time on Tuesday, March 24, 2026. Responses should be emailed to the Contracting Officer, Manish Patel, at manish.patel@faa.gov. The email subject must be "ChBA TSS Request for Concept Paper." The FAA intends to send a confirmation email within 24 hours of receipt; an offeror not receiving a confirmation email should follow up with the Contracting Officer to confirm receipt.

Offerors may submit questions regarding this RCP to the Contracting Officer at manish.patel@faa.gov until 2:00PM Eastern Time on Tuesday, March 3, 2026.

The cover sheet and concept paper must be separate Adobe .pdf files named “RFP_CovSheet_[Offeror Name].pdf” and “RFP_Concept_[Offeror Name].pdf”. The cover sheet file must be formatted to letter-size paper with one-inch margins. Color is permissible, but each document should be readable if printed in black-and-white. All files must be submitted in the same email transmittal.

The Cover Sheet.pdf file and the Concept Paper.pdf file should contain only the number of pages allowed in the instructions above. Documents that exceed the page limitations will only be reviewed through the stated page limitation, which will result in the offeror’s full concept not being evaluated fully and eliminating progression to the next phase. The email transmittal may include additional text (such as whom to contact if the files are corrupted), but those emails will not be read by the peer review panel.

An offeror may submit multiple concept papers covering the FAA’s objective for this announcement. Each should be a separate email submittal following all instructions above. Multiple concepts should be submitted only where the concepts differ greatly in innovation or process. The FAA desires unique and innovative concepts. If multiple offerors submit very similar concept papers, the FAA or its designated agent may select any one or more of those for future phases of this challenge.

4. Evaluation Approach

The goal of the TSS E2 source selection process is to select the proposed solution that meets the FAA’s requirements and offers the best value to the FAA. The goal of the TSS E2 Phase I evaluation of concept papers is to identify offerors that may be invited to participate in future phases of the TSS E2 ChBA. Offerors will be eliminated from further participation in the TSS E2 ChBA if the FAA or its designated agent determines during the Phase I evaluation that it is unlikely that the offeror’s proposed solution would offer the best value to the FAA.

The FAA or its designated agent will evaluate responses by assessing the degree to which the response presents a complete and realistic solution that addresses all the objectives in the SOO and the requirements in this RCP and by assessing the totality of the strengths, weaknesses, and risks identified in the response. The definitions of strength, weakness, and risk are listed in Table 1.

Table 1: Definitions of Strength, Weakness, and Risk

Strength	An aspect of an offeror’s concept that exceeds the requirements and, as a result, increases the probability of successful performance.
Weakness	An aspect of an offeror’s concept that fails to meet a requirement or provides a demonstrable or reasonably supported reduction in the likelihood of successful performance. An element will be rated a weakness if it meets one or both criteria.

Risk	An aspect of an offeror's concept that, when assessed as to its probability of occurrence and the magnitude of its impact, presents an uncertainty as to the ability of the offeror to successfully perform the solution.
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4.1 Technical Approach and Architecture

Concept papers will be evaluated based on:

- The degree to which the TSS solution provides high fidelity simulation capability for the 264 towers in the NAS.
- The degree to which the TSS solution provides for rapid video database (VDB) facility file modification, and rapid access remotely. (SOO VDB)
- The degree to which the TSS solution provides for rapid training scenario development and deployment. (SOO scenario development)
- The degree to which the TSS solution provides for a realistic and high-fidelity virtual pilot representation, emulating natural speech in varying accents. (SOO Virtual Pilot)
- The degree to which the TSS solution provides an effective approach to integrating remote instructors into the TSS simulation experience. (SOO Remote instructor)
- The degree to which the solution provides an effective approach to providing feedback to air traffic controllers using the simulator and training record retention and retrieval for use by remote evaluators. (SOO automated evaluation)
- The degree to which the offeror promotes innovative solution architectures that leverage advancements in the industry (e.g., open source, cloud computing) and modern offerings (e.g., no code/low-code, DevSecOps principles) and incorporates the use of artificial intelligence (AI) to reduce the reliance on human interaction. (SOO: Architecture Objective)
- The degree to which the offeror addresses FAA safety and cybersecurity requirements. (SOO: Security Objective)
- The degree to which the offeror supports industry-standard capabilities for real-time monitoring, alerting, and reporting of health, performance, and security status. (SOO: Monitoring Objective)
- The degree to which the TSS solution provides remote TSS software updates without interference with ongoing usage or training in progress.
- The degree to which the TSS solution provides real-time access to usage data for each simulator, scenario, and simulated airport by location to enable the FAA to monitor and

evaluate the performance of the TSS solution.

4.2 Implementation Approach

Concept papers will be evaluated based on:

- The viability of the implementation plan to achieve installation of working TSS systems at 102 sites by May 2028.
- The degree to which the implementation plan effectively addresses assumption and risks to achieving deployment by May 2028.

4.3 Operational Support Approach

Concept papers will be evaluated based on:

- The degree to which the offer provides a comprehensive cost-efficient approach to life cycle support.
- The degree to which the offer provides an effective and efficient approach to on-call and in person training throughout the life cycle.

4.4 Business Viability

Concept papers will be evaluated based on:

- The degree to which the offeror provides a reasonable ROM estimate.
- The viability of the overall estimate to achieve the results desired by the Government.

5. Disclosure of Information

Non-government personnel will be used in the evaluation of the concept papers. The non-government advisors may have access to all aspects of the offeror's submitted materials. By submitting a concept paper, your company agrees with the use of non-government advisors.

Please do not submit any classified information.